

Exploring the Spatial Distribution of Persistent SARS-CoV-2 Mutations - Leveraging mobility data for targeted sampling

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The authors analyze the relationship between human mobility and genomic data of SARS-CoV-2 using mobile phone mobility data and sequence data and present a **solid** proof of concept. This **useful** work was conducted on a fine spatial scale and provides suggestions on how mobility-derived surveillance could be conducted, although these results are mixed. The primary significance of this work is the strong use of large datasets that were highly granular. The authors provide a rigorous study, but with less clear predictive power of mobility to inform transmission patterns.

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Abstract

Given the rapid cross-country spread of SARS-CoV-2 and the resulting difficulty in tracking lineage spread, we investigated the potential of combining mobile service data and fine-granular metadata (such as postal codes and genomic data) to advance integrated genomic surveillance of the pandemic in the federal state of Thuringia, Germany. We sequenced over 6,500 SARS-CoV-2 Alpha genomes (B.1.1.7) across seven months within Thuringia while collecting patients' isolation dates and postal codes. Our dataset is complemented by over 66,000 publicly available German Alpha genomes and mobile service data for Thuringia. We identified the existence and spread of nine persistent mutation variants within the Alpha lineage, seven of which formed separate phylogenetic clusters with different spreading patterns in Thuringia. The remaining two are sub-clusters. Mobile service data can indicate

these clusters' spread and highlight a potential sampling bias, especially of low-prevalence variants. Thereby, mobile service data can be used either retrospectively to assess surveillance coverage and efficiency from already collected data or to actively guide part of a surveillance sampling process to districts where these variants are expected to emerge. The latter concept was successfully implemented as a proof-of-concept for a mobility-guided sampling strategy in response to the surveillance of Omicron sublineage BQ.1.1. The combination of mobile service data and SARS-CoV-2 surveillance by genome sequencing is a valuable tool for more targeted and responsive surveillance.

Introduction

On March 11, 2020, the World Health Organization (WHO) classified the SARS-CoV-2 virus (Severe Acute Respiratory Syndrome - Corona Virus - 2) as a global pandemic due to its rapid spread and high infection rate.¹ The airborne virus has since caused significant morbidity and mortality worldwide (<https://covid19.who.int/>). In an attempt to control its spread, many countries initiated comprehensive surveillance efforts with molecular techniques such as Polymerase Chain Reaction (PCR) and Whole Genome Sequencing (WGS).^{2,3} Consequently, nearly 15.8 million SARS-CoV-2 sequences have been deposited into the "Global Initiative on Sharing All Influenza Data" database (as of July 21, 2023, GISAID). Many research groups have undertaken studies examining the viral spread by integrating sequencing and epidemiological data to monitor the pandemic and investigate local outbreaks.^{4,5} Most of these local projects are part of national surveillance programs such as the UK's Genomics Consortium (COG-UK) or "national genomic surveillance" in the USA.^{2,6} In Germany, the "Coronavirus-Surveillanceverordnung" (CorSurV) enacted by the State Ministry of Health on January 19, 2021, mandated that laboratories with sequencing capabilities process SARS-CoV-2-positive samples, offering financial compensation until April 2023.³

Bioinformatics workflows developed in Germany, such as poreCov (for Oxford Nanopore data) and CoVpipe2 (for Illumina data), reconstruct SARS-CoV-2 consensus genomes from the sequencing data and prepare the results for upload and submission to the Robert Koch Institute (RKI).^{7,8} As the German government's public health and biomedical research institute responsible for disease control and prevention, the RKI collected the genomes via the German Electronic Sequence Data Hub (DESH) and integrated them with additional epidemiological information to provide an up-to-date overview of the ongoing viral spread. For keeping track of the rapid SARS-CoV-2 evolution, PANGO (Phylogenetic Assignment of Named Global Outbreak) provides a standard naming convention based on unique mutation profiles and further criteria, resulting in the classification of over 3,660 lineages (as of August 2023).^{9,10} Additionally, the WHO classified important viral lineages as "Variants of Concern" (VOC), "Variants of Interest" (VOI), or "Variants under Monitoring" (VUM), using Greek designations in the past (e.g., "Alpha" (Pango lineage main designation B.1.1.7) or "Omicron" (Pango lineage main designation B.1.1.529)). Further, the WHO also de-escalated former VOCs to reflect the current SARS-CoV-2 variant landscape better. The first defined VOC (now de-escalated), the Alpha lineage, rapidly replaced almost all previously circulating lineages globally by the end of 2020 until the VOC Delta (main lineage B.1.617.2) replaced it in mid-2021.¹¹⁻¹³

To predict or monitor the rapid viral spread throughout regions, various data types, like travel data, passenger volumes, or passive wastewater monitoring were examined previously.¹⁴⁻¹⁶ Furthermore, different studies explored mobility data with genomic data to retrace the origin and spatial expanse of Alpha or utilized geolocation data to model the spread in metropolitan areas to recreate case trajectories and the impact of mobility restrictions.^{17,18} Mobility data was also used in Germany during the pandemic, revealing that lockdowns leave distant parts of the country

less connected due to the sharp decline in long-distance travel.¹⁹ These studies focused on analyzing residential movement and contact tracing to evaluate and inform health policies but were not applied to active molecular surveillance.

Here, we investigated whether mobile service data and fine-granular metadata (such as postal codes and genomic data) can help predict the spread of the Alpha lineage or guide the sampling for more targeted genomic surveillance with a focus on the German federal state of Thuringia.

Results

The Alpha lineage spread rapidly through Thuringia, showing a pattern similar to its nationwide spread

Thuringia is a rural federal state in central Germany with a population of 2.1 million and no major airports (overview of Thuringia's population density in Supplementary Figure S1). We investigated if the spread of the Alpha lineage of SARS-CoV-2 behaved differently compared to the whole of Germany. To understand its spread, we used 289,487 public SARS-CoV-2 genomes from Germany (excluding Thuringia; including 137,024 Alpha genomes) and 7,394 genomes from our own sequencing data for Thuringia (including 6,307 Alpha genomes) to track Alpha's occurrence from December 2020 to August 2021 (see **Figure 1**, Supplementary Figure S2, and Supplementary Tables S1 and S2; for details, see Methods section "Alpha spread datasets"). For Thuringia, district-level data (full postal code) per genome were available, whereas, for Germany, only postal code data of the sending laboratories (referred to as "primary diagnostic laboratory" by the RKI where the SARS-CoV-2 positive sample was detected) and sequencing laboratories were publicly available.

In late December 2020, six federal states in Germany (from here on called states) reported the first cases of the Alpha variant. Although sequencing was initially low, it gradually increased in the following month. However, the Corona-Surveillance regulation was passed at the end of January 2021, leading to a rapid increase in sampling and sequencing by February since sequencing costs could be reimbursed. Even though Thuringia sequenced a similar amount of SARS-CoV-2 samples compared to other German states (as shown in **Figure 1**), the proportion of the Alpha variant to other lineages was relatively low. However, the proportion of Alpha increased heavily in February.

By March, Alpha had spread to nearly all states and districts (districts are similar to counties or provinces) in Germany (Median: 76.47 % Alpha samples among a federal states total sequenced samples compared to 36.03 % in February, excluding Thuringia) and Thuringia (Median: 85.29 %, up from 50.00 % in February). So, there was no noticeable difference in the Alpha proportions between Germany and Thuringia after February. During the summer of June and July 2021, sequencing declined in Germany (including Thuringia; Supplementary Figure S2) due to the decrease in overall daily cases, as reported by Meintrup *et al.* and Oh and Hölzer *et al.*^{20,21}

In summary, the spread of the Alpha lineage in Thuringia lagged roughly two weeks behind the general spread of other German federal states but showed similar proportions. This suggests that Thuringia experienced a delay in the initial arrival of Alpha. However, we did not observe any difference in the overall spread afterward. Thuringia was among the first states to adopt new containment measures, including contact limitations, closure of retail shops, and prohibition of tourist journeys (14th December 2020). Jena, a city in Thuringia, was also the first German city to implement mandatory public masking in March 2020.²² Contacts were further restricted on January 9th, and people were urged to restrict their movement radius to 15 km, which might explain the delay besides the absence of major airports nearby.

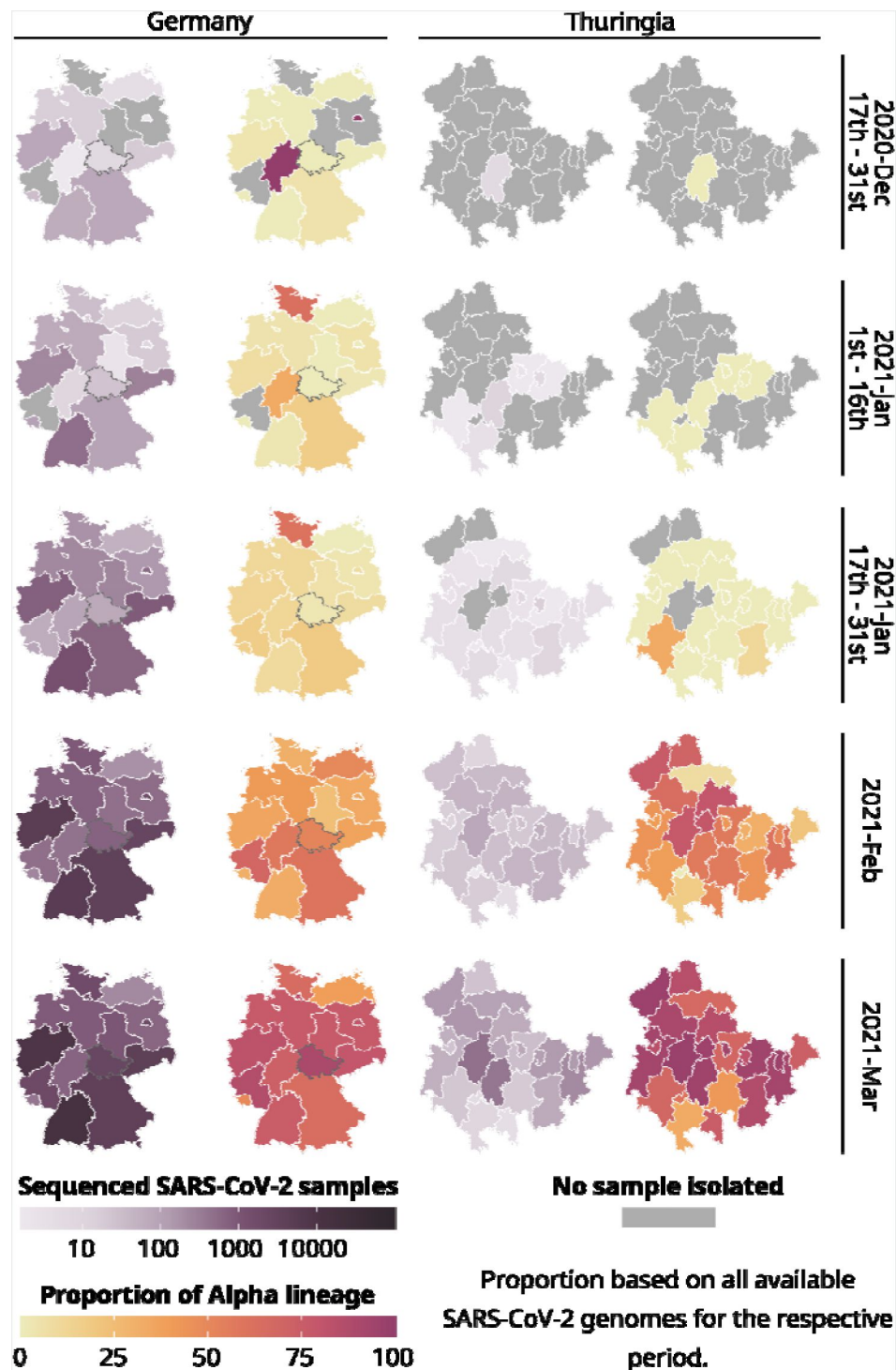


Figure 1

Total number of all sequenced SARS-CoV-2 samples (purple) and the proportion of the Alpha lineage for all sequenced samples (yellow-red) for each state of Germany and each district of Thuringia.

289,487 publicly available German SARS-CoV-2 genomes and their metadata were used for the general German maps, excluding data from Thuringia. For Thuringia, we always used 7,394 genomes and their metadata from our database for the German and Thuringian maps. Please note that for all states except Thuringia, we used the postal code of the sending laboratory as a proxy for the geographic location of a sample. A gray border on the maps of Germany highlights Thuringia.

All Thuringian genomes were evenly distributed between other German samples in the phylogenetic time tree (see Supplementary Figure S3). However, due to its rapid spread from February onwards, it is difficult to accurately track how the Alpha lineage specifically expanded (point of entries, exact origins, etc.). Consequently, we investigated whether "sublineages" might be identifiable and trackable to address this.

Monitoring of Alpha subclusters in Thuringia reveals temporally and regionally restricted distribution patterns

To identify possible clusters among the Alpha lineage spreading in Thuringia, we called each Alpha genome's mutations via Nextclade by analyzing them using poreCov^{7,23}. We identified nine clusters out of 70,429 Alpha genomes, based on their mutation profile, time period, and phylogenetic distance (from here on called Alpha subclusters; for details, see Methods "Subcluster identification"). All subclusters, their time period, and sample size in Thuringia are summarized in **Table 1**. An overview of each subcluster (phylogenetic time tree, location, and period) is also provided on *microreact.org* as interactive views (see Methods "Subcluster identification"). Note that our subcluster definition is similar to the definition of a sublineage. However, PANGO sublineages were rarely defined during the Alpha wave (PANGO designation: Q.1 to Q.8; compared to the Delta and Omicron waves).

Eight of these subclusters are based around a specific spike protein mutation, while the other contains a mutation within the ORF1b region. The subcluster 7.1 "S:N185D" branched out from the subcluster 7 "ORF1b:A520V" and subcluster 6.1 "S:V90F" is a branch of subcluster 6 "S:S939F" (see **Table 1**). These two branched subclusters still carry the specific mutation of their originating subcluster. The subclusters 3, 4, and 5 were observable between two and three months, and the other subclusters over at least four months. To investigate these subclusters' regional spread, each sample was mapped to its Thuringian district based on the resident's postal code from which it was isolated. We then sorted the samples according to their subclusters and visualized them throughout the subcluster's observed period. The spread of two representative subclusters is exemplary visualized in **Figure 2a**, and all the subclusters are available via Supplementary Figure S4 and their data in Supplementary Table S3. Additionally, all subclusters and their metadata are also available via Microreact (see Methods "Subcluster identification").

Each of the seven main mutation variant clusters originated from a different Thuringian district. At the same time, two subclusters, 6.1 and 7.1, branched out from the same districts as their original clusters (6 and 7) 12 or 13 days after their first emergence. The subclusters mainly spread regionally confined and not across all of Thuringia (see **Figure 2a** and Supplementary Figure S4) but were also identified in other states of Germany (see "*microreact.org*"-project). For example, the "S:S939F" subcluster spread across 15 states, with the first samples being isolated outside of Thuringia. The eight Spike-mutation subclusters had expanded between four to twelve of the 23 Thuringian districts within the observable time period of each subcluster. They expanded by one to six districts per month, with a greater expansion accompanied mainly by a larger increase in the subcluster sample number. In contrast, the ORF1b-variant even comprised 21 districts and expanded between two to seven districts per month (see **Figure 2a**). Most of each subcluster's samples were identified in their region of first occurrence, and no additional samples were found after the given periods.

Several limitations need to be considered. The identified subclusters may have multiple origins or may not originate from Thuringia. Due to the lack of precise zip codes (publicly available German genomes are limited to postal codes of sending and sequencing laboratories), monitoring the subclusters in other states on a district level was impossible. Nevertheless, we could follow how the subclusters developed in Thuringia, even if multiple origins may have affected the overall speed or length of each subcluster's occurrence.

Designation	Mutations	Number of samples	Time period	Remarks
1	S:H49Y , ORF1a:I841V	44	Feb-May 2021	S:H49Y eases cell entry in S-pseudotyped lentiviral system. ²⁴
2	S:N354K	63	Feb-May 2021	S:N354K slightly impaired mAb h11B11. ²⁵
3	S:G496S , ORF1a:E1013K	12	Mar-May 2021	S:G496S: compromises BA.1 replication fitness and changed mAb sensitivities, reduces ACE2 binding affinity, and increases immune evasion. ²⁶⁻²⁸
4	S:N703D , ORF1a:D1228G, ORF1a:A2123V	51	Mar-May 2021	-
5	S:T716V , N:G204P, ORF1a:D1600N	22	Apr-May 2021	-
6	S:S939F	206	Feb-May 2021	S:S939F: modulates T-cell propensity. ²⁹
6.1 [#]	S:V90F , S:S939F	55	Feb-May 2021	-
7	ORF1b:A520V	811	Feb-Jun 2021*	-
7.1 ^{\$}	S:N185D , ORF1b:A520V, ORF1b:L1504F	40	Feb-May 2021	-
* only one sample for June # branch from subcluster 6 \$ branch from subcluster 7				

Table 1

Overview of nine Alpha subclusters in Thuringia, their sample count, their time period, and their specific mutations that are shared across all members of the subcluster (excluding characteristic Alpha mutations that are shared across all subclusters). The mutation used to define the subcluster is highlighted in bold.

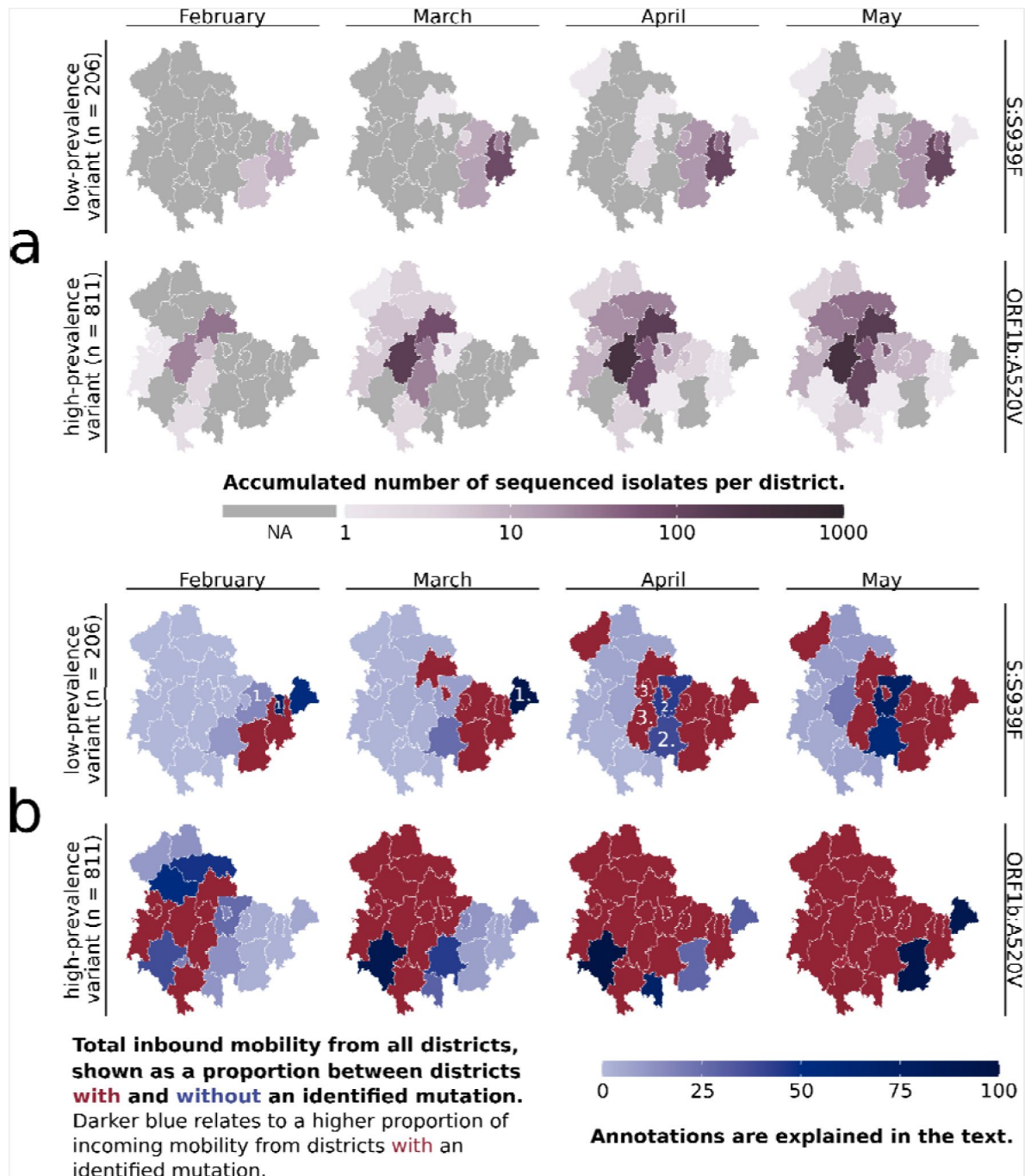


Figure 2

Overview of the subclusters "S:S939F" and "ORF1b:A520V" in Thuringian districts.

a) Accumulated number of sequenced samples for each subcluster per district and per month. b) Combined visualization of each district's "inbound mobility" from other districts (color intensity) and the occurrence of a subcluster sample (red = sample found, blue = no sample found). The inbound mobility of each district (blue color intensity) is shown as a proportion of incoming mobility from other districts with or without an identified sample. The darker the blue color of a district, the higher the proportion of inbound mobility from other districts with an identified subcluster sample (red districts). The light blue color describes that most of the inbound mobility of a district comes from other districts without an identified subcluster sample (blue districts). Numbers refer to district types 1, 2, and 3, as further defined in the main text. The last month of subcluster "ORF1b:A520V" is not visualized, as affected districts were unchanged.

Our surveillance sampling heavily relies on various institutions and partners, and only a portion of the provided samples can be sequenced (see “Sampling” in Methods). For example, the spread of subcluster “S:S939F” revealed two districts in April where no respective samples were found (**Figure 2a**) despite being surrounded by districts with “S:S939F”-samples present. This could be due to the lack of samples sent to sequencing from those regions or the low prevalence. We, therefore, investigated if mobile service data of residents, in addition to molecular surveillance, might be utilized to counteract this issue.

Mobile service data indicates Alpha subcluster spread and sampling bias

With the aim to predict the subcluster spread and, thereby, reduce surveillance-based sampling bias, we utilized anonymized mobile service data from T-Systems International GmbH. Around 200 million trips were used to determine the number of daily trips between the Thuringian districts. We then combined this information with our fine-granular genomic data to specify each district’s monthly proportion of inbound mobility from subcluster-affiliated districts (see Methods “Mobile service data”). The results are visualized in **Figure 2b** (complete overview in Supplementary Figure S5; data provided in Supplementary Table S4).

The mobile service data-based prediction of a subcluster’s spread aligned well with the subsequent regional coverage of fast-spreading, highly prevalent subclusters, such as subcluster 7, which covered 811 samples (see **Figure 2**). In contrast, the predicted spread for the low-prevalence subclusters did not correspond well with the actual occurrence. Yet, adding mobile service data resulted in three different “types” of districts (see **Figure 2b**, annotated districts). Type 1 included districts with high inbound mobility from areas with an identified variant, where the variant was eventually found afterward, while Type 2 included districts with high inbound mobility from areas with an identified variant, where the variant was never identified. Type 3 included districts not directly connected to a district with an identified variant, but a variant was eventually identified while they border Type 2 district(s). Our previous analysis of the subclusters’ spreading pattern across the districts, based solely on identified variants, indicated missed identifications in some districts due to the seemingly illogical spread to districts without a connection to others (**Figure 2a**, subcluster “S:S939F”). The inclusion of mobile service data revealed some of these districts to be Type 2 districts. This suggests that the specific variant should be identifiable within these districts due to the observed high incoming mobility from districts with identified variants. Type 2 districts were mainly observed for subclusters with low prevalence and, consequently, low numbers of covered samples that are usually more difficult to monitor. For example, we assumed missing identifications in some districts of subclusters 1, 2, and 3, which, through the mobile service data, are now partially identified as Type 2 districts. However, due to their low prevalence, it is also possible that these subclusters have not spread to the indicated districts.

Despite analyzing the mobile service data of districts from other federal states than Thuringia, we could not apply them, as the lack of precise location data for samples outside of Thuringia prevented the correct calculation of the incoming mobility. Based on the nine observable clusters, we concluded that mobile service data is a good prediction marker for the spread of high-prevalence variants but, more importantly, a good indication of districts that should have an identified low-prevalence variant. Next, we investigated if mobile service data can improve active surveillance via guiding sample collection for genomic sequencing.

Proof of principle: Mobile service data-guided sampling for genomic surveillance for Omicron BQ.1.1

Based on our previous findings, we implemented the “mobility-guided” sampling approach under real pandemic circumstances over one month in addition to our active surveillance.

As the subject of investigation, we searched for a newly emerging (based on global news reports) and ideally low prevalent SARS-CoV-2 lineage in Thuringia.

Among the various emerging Omicron sublineages during that time, sublineage BQ.1.1 fulfilled the defined criteria. First isolated in a northwestern Thuringian community with around 20,000 inhabitants on October 5, 2022, we identified this particular sublineage on October 14, 2022, among a routine batch of 42 samples. BQ.1.1 was a low-prevalence sublineage that was identified worldwide (<https://outbreak.info/situation-reports?pango=BQ.1.1> )

Following its first Thuringian identification, we utilized the latest available dataset of the past two years of mobile service data (October 2020 and June 2021) to investigate the residential movements for the community of first detection. Considering the highest incoming mobility from both datasets, we identified 18 communities with high (> 10,000), 34 with medium (2,001-10,000), and 82 with low (30-2,000) number of incoming one-way trips from the originating community (purple triangles in **Figure 3a** )

. As a result, we specifically requested all the available samples from the eight communities with the highest incoming mobility. Still, we were restricted to the submission of third parties over whom we had no influence. This led to the inclusion of the following eight communities with the most residential movement from the originating community: four in central and three in NW of Thuringia, one in NW-neighboring state Saxony-Anhalt. The samples requested from central Thuringia were also due to their geographic arrangement as a “belt” in central Thuringia, linking three major cities (see Supplementary Figure S1). Subsequently, we collected 19 additional samples (isolated between the 17th and 25th of October 2022; see “Guided Sampling” for October 2022, **Figure 3a** ) besides the randomized sampling strategy. Thus, the sampling depth was increased in communities with high incoming mobility from the first origin.

As part of the general Thuringian surveillance, we collected 132 samples for October (covering dates between the 5th and 31st) and 69 samples in November (covering dates between the 1st and 25th; see **Figure 3b** ) and **3c** )

. Randomized sampling was not influenced or adjusted based on the mobility-guided sample collection. Thus, it also contains samples from communities with a mobility link towards the first occurrence of BQ.1.1, as they were part of the regular random collection (see gray triangles in **Figure 3b** ). A complete overview of all samples is provided in Supplementary Table S5. The mobility datasets from October 2020 and June 2021 for all sampled communities are provided in Supplementary Table S6.

Among the 19 samples specifically collected based on mobile service data, we identified one additional sample of the specific Omicron sublineage BQ.1.1 in a community with high incoming mobility ($n = 14$, number of trips = 37,499) with a distance of approximately 16 km between both towns. Our randomly sampled routine surveillance strategy did not detect another sample during the same period. This was despite a seven times higher overall sample rate, which included 31 samples from communities with an identified incoming mobility from the community of the first occurrence (October 2022, **Figure 3b** )

. Only in the one-month follow-up were four other samples identified across Thuringia through routine surveillance (November 2022, **Figure 3c** ).

During our attempt to implement the mobility-guided sampling approach in real-time during the pandemic, we encountered three distinct limitations, some of which are commonly observed in surveillance practices. The guided sampling depended on the individual sample submitting institutions, affecting the availability of suitable samples, especially for the communities of interest. By choosing a newly emerging Omicron sublineage for our experiment, spread and, therefore, suitability were uncertain. In our case, BQ.1.1's prevalence in Thuringia was even lower than expected, and it also remained rare in subsequent months, with only 42 samples found until June 2023, eight months after the first occurrence in Thuringia. Due to the short preparation time,

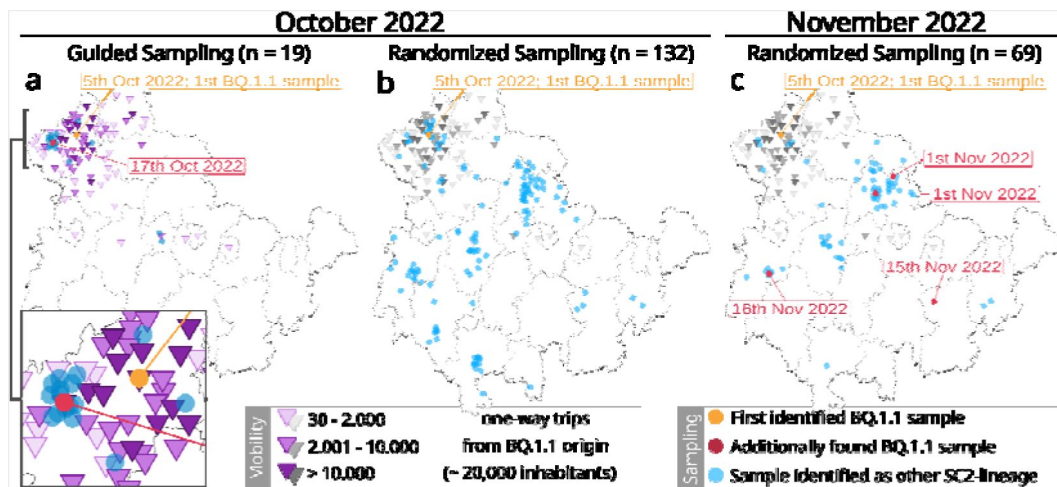


Figure 3

Overview of the mobility-guided sampling of the Omicron sublineage BQ.1.1 in Thuringia (a) compared to the default randomized sampling (surveillance) in October (b) and November 2022 (c).

The randomized surveillance results in November 2022 (c) have been added to highlight the spreading progress of BQ.1.1. Dots reflect the location of each sample (based on residents' zip codes). Orange dot: First identified BQ.1.1 sample; Red dot: Additionally identified BQ.1.1 sample; Blue dot: Another SARS-CoV-2 lineage. Purple triangles represent the number of one-way trips a community received from the community of the first BQ.1.1 occurrence (orange dot) based on mobility data from October 2020 and June 2021. The same mobility data from mobility-guided sampling (a) were added in grayscales to the randomized surveillance (b and c) as a visual reference only. Sampling dots are slightly scattered to improve visibility.

only mobile service data from the past two years and no current data were available. Nevertheless, the available datasets still reflect pandemic movement behavior since the pandemic has been ongoing for two years.

In summary, increasing the sampling depth in the suspected regions successfully identified the specified lineage using only a fraction of the samples from the randomized sampling. Conversely, randomized surveillance, the “gold standard” acting as our negative control, did not identify additional samples with similar sampling depths in regions with no or low incoming mobility or even in high mobility regions with less sampling depth. Implementing such an approach effectively under pandemic conditions poses difficult challenges due to the fluctuating sampling sizes. Although the finding of the sample may have been coincidental, our proof of concept demonstrated how we can leverage the potential of mobile service data for targeted surveillance sampling.

Conclusion

During the SARS-CoV-2 pandemic, diverse data sources like travel, wastewater, and mobility data have been employed in surveillance and transmission tracking.^{14,19} In the present study, we analyzed over 296,800 German SARS-CoV-2 genomes to examine whether mobile service data can predict the spatial distribution of the Alpha lineage in the German state of Thuringia and how they potentially benefit pandemic surveillance.

A plausible explanation for the delayed spread of the Alpha lineage in Thuringia is the lack of major transport hubs, as Alpha first occurred in federal states with such hubs. Previous studies have already highlighted the impact of major transportation hubs in the spread of Sars-CoV-2.^{14,30} However, its impact on the total distribution is limited, and the spread was ultimately comparable between Germany and Thuringia. While our findings on mobile service data may also apply to Germany, we could not verify this because the limited location data of publicly available German genomes prevented in-depth investigations outside of Thuringia. Thus, precise sampling location data are crucial to utilize mobile service data in genomic surveillance, but privacy regulations may restrict access to this data. Shortly after Alpha’s emergence, mutation variants formed like the known sublineages Q.1 to Q.8 and the Thuringian subclusters identified by us. This reflects the ongoing evolution during active circulation and indicates an even greater sublineage diversity, which has not been surveyed as closely as in the subsequent Delta and Omicron waves. By monitoring the nine Thuringian subclusters, rather than focusing solely on the parental lineage B.1.1.7, we were again able to effectively track transmissions and gain a comprehensive understanding of the regional spread. So, it underscores the importance of sequencing in pandemic surveillance to explore such genomic changes and, thereby, keep track of the transmission chains and potential outbreaks.

Mobile service data can support such surveillance in different ways. Previous studies examined the capabilities of mobility data in the context of, e.g., case trajectories, but retrospectively applied to already collected data, it can be used to examine surveillance sampling coverage and possible sampling bias. We exemplified this approach with the Alpha lineage, where mobile service data indicated a putative sampling bias and partially predicted the spread of our Thuringian subclusters.

Another approach is actively guiding the sampling process through mobile service data, which we demonstrated with our proof of principle focusing on the Omicron-lineage BQ.1.1 as a real-life example. This approach could allow for a flexible allocation of surveillance resources, enabling adaptation to specific circumstances and increasing sampling depth in regions where a variant is anticipated. By incorporating guided sampling, much fewer resources may be needed for unguided or random sampling, thereby reducing overall surveillance costs.

Additionally, while this approach is particularly useful for identifying low-prevalence variants, it is not limited to such variants. Still, it can provide a guided, more cost-efficient, low-sampling alternative to general randomized surveillance that can also be applied to other viruses or lineages. For this purpose, Pre-generated mobility networks automatically tailored to each state's unique infrastructure and population dynamics could provide better-targeted sampling guidance rather than simple geographical proximity. However, the feasibility depends on the availability and cost of such mobile service data. Alternatively, financial resources could also be invested directly in increasing sampling capacity and coverage, which ultimately depends on individual factors of the respective surveillance. Mobile service data can also be used with other surveillance approaches and elements. For example, wastewater surveillance can give further indications to supplement guided sampling. At the same time, passenger data offers additional insights into traffic hubs as sources of regional movement.

Methods

Sampling

Starting mid-2020, we initially sequenced hospital-intern samples, transitioning by January 2021 to approximately 43 PCR-positive samples per week: 20 from the hospital's microbiology department and 23 randomly sourced by the Thuringian State Authority for Consumer Protection ("Thüringer Landesamt für Verbraucherschutz"; TLV).

Until June 2023, our institute sequenced 3,770 SARS-CoV-2 samples, and SYNLAB Holding Deutschland GmbH, Bioscientia Healthcare GmbH, and DIANOVIS GmbH provided additional 7,800 Thuringian SARS-CoV-2 genomes and their metadata.

Sample preparation and sequencing

RNA isolation used the ZymoResearch "Quick-RNA Viral Kit" (Zymo Research Europe GmbH, Germany, Product-ID: R1035), according to the manufacturer's instructions with 100 µl patient sample input and a centrifuge speed of 16,000 g.

The viral RNA underwent a Reverse Transcriptase (RT)-PCR followed by a multiplex-PCR using the ARTIC V1200 primer set, according to Freed and Silander's SARS-CoV-2 sequencing protocol (version 4, updating to version 5 by March 2021).³¹ Subsequent DNA quantification utilized the Qubit dsDNA HS assay (Invitrogen, USA).

From the amplified DNA, a sequencing library was prepared using the Nanopore SQK-LSK109 and SQK-RBK004 kits (Oxford Nanopore Technologies, Oxford, UK), sequenced for a maximum of 72 h utilizing an Oxford Nanopore MinION Mk1b sequencer with R.9-flowcells and the MinKNOW software (versions MKE_1013_v1_revBC_11Apr2016 to MKE_1013_v1_revBR_11Apr2016 in the respective period), and analyzed with the software pipeline poreCov (versions 0.3.5 to 0.11.7; including basecalling, demultiplexing, adapter removal, quality filtering, and genome alignment) to reconstruct consensus genomes.⁷ Sequencing data and the respective metadata (e.g., isolation date, sending laboratory details) were submitted to the RKI through DESH. We also collected the postal code of the isolation location or at least of the sending local health authority, storing all data additionally in a local database.³² Due to data protection, such data is limited on the RKI's public GitHub repository (https://github.com/robert-koch-institut/SARS-CoV-2-Sequenzdaten_aus_Deutschland), providing instead postal codes of the sequencing and sending laboratories.

Alpha spread datasets

From our local database, we extracted 8,397 samples with isolation dates before Oct 1st, 2021. After adding federal state and district information, 993 entries with non-Thuringian locations were excluded. Further, ten entries with unspecific isolation dates were excluded, yielding 7,394 samples (including 6,307 Alpha genomes (lineages B.1.1.7 and Q.1 to Q.8)). The publicly available RKI SARS-CoV-2 dataset was downloaded, containing 1,091,655 genomes with the respective metadata (17th Oct 2022; Zenodo-version 2022-10-16).³³ 789,405 entries, isolated after Sep'21, and 59 entries without “sending laboratory” information were removed. For the resulting 302,191 entries, location information (location, federal state, district, longitude, latitude) were added based on the sending laboratory postal code. Five entries with a non-existing postal code and all 12,704 Thuringian samples were removed from the dataset, resulting in 289,487 samples (including 137,024 Alpha genomes). We investigated only Alpha-lineage samples collected from September 2020 onwards, after the first official reports of the Alpha-lineage.¹¹

Analyzing both datasets, we calculated the monthly proportion of Alpha lineage samples in Thuringia and Germany per state/district, dividing Dec'20 and Jan'21 into first and second halves.

Subcluster identification

Using a total of 70,429 German and Thuringian Alpha genomes, a phylogenetic time tree was created (see Supplementary Method “Phylogenetic time tree construction” and Supplementary Figure S3). We determined the frequency of all non-Alpha-specific mutations among the 6,522 Thuringian Alpha genomes. We then manually screened for mutations present in at least 20 genomes with a small phylogenetic distance and a time occurrence of at least two months. This led to nine mutations, each of them creating a defined cluster covering between 12 and 811 closely related genomes. We only kept mutation information of these nine subclusters in the respective metadata, which, together with the tree file of the phylogenetic time tree, was uploaded to a “microreact.org”-project, provided as Supplementary File 1 and found under the following link: <https://microreact.org/project/ftR2GjfF6iXtSwbmN4ARTx-thuringianalpha-linclusters#76ir-complete-overview>.

Mobile service data

T-Systems International GmbH collected and aggregated mobile service data via the Cell ID method, dividing a geographical area into so-called traffic cells. Each cell is assigned to exactly one transmitter mast, with a spatial resolution from 500 m x 500 m up to 8 km x 8 km (depending on the transmitter mast network density). Cell phones always register to the closest traffic cell, which is recorded and stored in an Origin-Destination Matrix (ODM). For population representation, the data was extrapolated with Deutsche Telekom's market share. Due to data privacy, the registration data is combined into movement streams between traffic cells, the status resolution is reduced to one hour (greater time intervals = less resolution), and individual traffic cells are grouped into districts. The degree of anonymization (k-value = 30, data privacy regulation) removed movement streams with less than 30 participants, resulting in approximately 200 Mio trips in the ODM. SMA Development GmbH analyzed all movements between the single Thuringian districts, adding each Alpha sample's isolation time and location data (per subcluster). The movements were further divided by months and originating district (subcluster-affiliated vs. -unaffiliated), determining each district's monthly inbound mobility proportion from cluster-affiliated districts.

Research in Context

Evidence before this study

We searched Pubmed for studies about the use of mobile service data for surveillance written in English. For the broadest possible search, we included any publication covering mobile data and surveillance aspects, using the following search string: ("cellular data" OR "cell phone data" OR "mobility data" OR "movement data" OR "migration data" OR "phone data") AND ("Surveillance" OR "Monitoring" OR "Survey" OR "Pandemic" OR "Disease" OR "Epidemic" OR "Outbreak"). Our search yielded 1,285 publications published between 1966 and 2023. We manually screened all these publications but found no study that applied mobile service data for active, targeted surveillance. Across all studies, the general focus was on tracking contacts or analyzing movements to assess, for instance, the efficiency of non-pharmaceutical interventions or generate prediction models. Some studies suggested targeted surveillance based on their results, but it was not yet applied. Additionally, we used "suite.ai" and "chatGPT" (with BING-search access) to let them search for "studies that utilize mobile service data to guide the sampling process for infectious disease surveillance". While "suite.ai" found two studies and "chatGPT" found another ten studies and reviews, none covered the direct application of the mobility data in active surveillance.

Added value of this study

This study highlights the value of combining mobile service data with fine-granular metadata for integrated genomic surveillance during the SARS-CoV-2 pandemic in a German federal state. We illustrated this strategy with the Omicron sublineage BQ.1.1 and how to guide the sampling processes toward areas where the new variant was expected to emerge. Additionally, we used mobile service data during the pandemic to assess our sampling coverage. Our study is the first to actively guide part of the genomic surveillance process during a pandemic.

Implications of all the available evidence

Efficient molecular surveillance setups are crucial in managing outbreaks from the local to the global scale. Different data sources are investigated to increase this efficiency, addressing factors like the more efficient usage of scarce surveillance resources and the prediction of spread. Extending molecular surveillance with such data should improve the future management of pandemics and outbreaks.

Additional information

Author contributions

Sample collection and preparation, R.Sp, R.Sc., M.L., and M.M.; sequencing, R.Sp., M.L., and M.M.; database setup and maintenance, M.J.; software, C.B.; bioinformatic analysis, R.Sp., M.L., M.M., and C.B.; mobile service data analysis, C.H., and M.Ha.; literature research, R.Sp.; writing first draft, R.Sp.; reviewing and editing manuscript, R.Sp., C.B., M.M., C.F.-S., A.K., C.H., M.Ha., M.Hö., D.K., R.Sc., P.D., and M.W.P.; supervision, C.B.; project administration, C.B.; funding acquisition, M.W.P.. All authors have read and agreed to the published version of the manuscript.

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Conflicts of interest

The authors have declared that no competing interests exist.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Grammarly in order to correct general English and improve readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Data Availability

All genomic data (genomes and respective metadata) are available in the provided microreact project (project file available under <https://osf.io/n5qj6/>; <https://microreact.org/project/ftR2GjfF6iXtSwbmN4ARTx-thuringianalpha-linclusters#76ir-complete-overview>). Supplementary tables and the R-scripts used to generate all figures are also provided in the repository under <https://osf.io/n5qj6/>. These include the mobile service data used in this study, which is available in processed and anonymized form. The original mobile service data can not be made public due to legal reasons/ownership. As the collector of the original aggregated mobility data, T-Systems applies to the EU- and German Data protection regulations.

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Reviewer #1 (Public review):

Summary:

In this manuscript Spott et al. combine SARS-CoV-2 genomic data alongside granular mobility data to retrospectively evaluate the spread of SARS-CoV-2 alpha lineages throughout Germany and specifically Thuringia. They further prospectively identified districts with strong mobility links to the first district in which BQ.1.1 was observed to direct additional surveillance efforts to these districts. The additional surveillance effort resulted in the earlier identification of BQ.1.1 in districts with strong links to the district in which BQ.1.1 was first observed.

Strengths:

There are two important strengths of this work. The first, is the scale and detail in the data that has been generated and analyzed as part of this study. Specifically, the authors use 6,500 SARS-CoV-2 sequences and district level mobility data within Thuringia. I applaud the authors for making a subset of their analyses public e.g. on the associated micro react page.

Further, the main focus of the article is on the potential utility of mobility-directed surveillance sequence. While I may certainly be mistaken, I have not seen this proposed elsewhere, at least in the context of SARS-CoV-2. The authors were further able to test this concept in a real world setting during the emergence of BQ.1.1 and compare it to the "gold standard" of random sampling. This is a unique real-world evaluation of a novel surveillance sequencing strategy and there is considerable value in publishing this analysis. Given the increased focus on optimizing sampling strategies for genomic surveillance, this work provides a novel strategy and will hopefully motivate additional modeling and real-world implementations.

Weaknesses:

The article is quite strong and I find the analyses to generally be rigorous. Limitations of the analysis, particularly due to the fact that BQ.1.1 remained a low-prevalence variant, are adequately addressed. The results do not provide quantitative, definitive proof that mobility-guided sampling is an optimal strategy, but they also do not claim to nor do I think they need to make an important contribution to the field.

<https://doi.org/10.7554/eLife.94045.2.sa2>

Reviewer #2 (Public review):

In the manuscript, the authors combine SARS-CoV-2 sequence data from a state in Germany and mobility data to help in understanding the movement of virus and the potential to help decide where to focus sequencing. The global expansion in sequencing capability is a key outcome of the public health response. However, there remains uncertainty how to maximise the insights the sequence data can give. Improved ability to predict the movement of emergent variants would be a useful public health outcome.

However, I remain unconvinced that changing surveillance strategies is necessarily sensible as it remains unclear what the ultimate benefit of variant hunting is. Decisions to adapt surveillance strategies should not be taken lightly as there are substantial benefits of maintaining a stable and as representative as possible, system over time. It's unclear what public health action would result of detecting a few more sequences of a variant. Once a variant has been identified (arguably anywhere in the world/region), we already have the necessary information to motivate the development of updated vaccines/monoclonals.

<https://doi.org/10.7554/eLife.94045.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Thank you for your assessment and constructive critique, which helped us to improve the manuscript and its clarity. Upon carefully reading through the comments, we noticed that, based on the Reviewer's questions, some of our answers were already available but "hidden" as supplementary data. Thus, we changed the following two figures and text accordingly to showcase our results to the reader better:

A) To highlight how mobile service data can indicate the spread of highly prevalent variants, we added a high-prevalence subcluster to Figure 2 (previously shown in Supplementary Figures S4 and S5) and, in exchange, moved one low-prevalence subcluster from Figure 2 back into the supplement. The figure is now showing a low and a high prevalent subcluster instead of two low prevalent subclusters.

B) Based on Reviewer 1's question about where samples were taken in regards to the mobility data from the community of the first identification (negative controls), we now highlight all the mobility data that was available to us in Figure 3 (as triangles) instead of just a few top mobility hits for both - mobility guided and random surveillance (serving as a negative control for the former). This way, we think, it is clearer how random sampling was also performed in some regions where mobility was coming from the community of origin (as asked by Reviewer 1) - the detailed trips and sampling are now part of the supplement for data transparency reasons. We also noticed a typo in the GPS coordinates, aligning one of the arrows falsely, which is corrected in the improved Figure 3.

We have also included the R-Scripts used to generate all the figures in the manuscript in an OSF repository (we updated the "Data sharing statement"). We also updated Figure 1 slightly and extended the supplemental material. The remaining comments to reviewers are addressed point-by-point below.

Reviewer 1 (Public Review):

In "1 Exploring the Spatial Distribution of Persistent SARS-CoV-2 Mutations -Leveraging mobility data for targeted sampling" Spott et al. combine SARS-CoV-2 genomic data alongside granular mobility data to retrospectively evaluate the spread of SARS-CoV-2 alpha lineages throughout Germany and specifically Thuringia. They further prospectively identified districts with strong mobility links to the first district in which BQ.1.1 was observed to direct additional surveillance efforts to these districts. The additional surveillance effort resulted in the earlier identification of BQ.1.1 in districts with strong links to the district in which BQ.1.1 was first observed.

Thank you for taking the time to review our work.

(1) It seems the mobility-guided increased surveillance included only districts with significant mobility links to the origin district and did not include any "control" districts (those without strong mobility links). As such, you can only conclude that increasing sampling depth increased the rate of detection for BQ.1.1., not necessarily that doing so in a mobility-guided fashion provided an additional benefit. I absolutely understand the challenges of doing this in a real-world setting and think that the work remains valuable even with this limitation, but I would like the lack of control districts to be more explicitly discussed.

Thank you for the critical assessment of our work. We agree that a control is essential for interpreting the results. In our case, randomized surveillance ("the gold standard") served as a control with a total sampling depth seven times higher than the mobility-guided sampling. To better reflect the sampling in regards to the available mobility data, we revisited Figure 3 and added all the mobility information from the origin that was available to us. We also added this information to the random surveillance to provide a clearer picture to the reader. This now clearly shows how randomized surveillance covered communities with varying degrees of incoming mobility from the community of first occurrences, thereby underlining its role as a negative control. We updated the manuscript to reflect these changes and included the October 2020 and June 2021 mobility datasets in Supplementary Table S6. We agree that the sampling depth increases the detection, which is the point of guided sampling to increase sampling, specifically in areas where mobility points towards a possible spread. In regards to the negative control: Random surveillance (not Mobility-guided) in October covered 40 samples in the northwest region of Thuringia (Mobility-guided covered 19 samples). Thus, random surveillance also contained 31 out of 132 samples with a mobility link towards the first occurrence of BQ1.1 but with varying amounts of mobility (low to high).

We added this information to the main text:

Line 270 to 293:

Following its first Thuringian identification, we utilized the latest available dataset of the past two years of mobile service data (October 2020 and June 2021) to investigate the residential movements for the community of first detection. Considering the highest incoming mobility from both datasets, we identified 18 communities with high ($> 10,000$), 34 with medium (2,001-10,000), and 82 with low (30-2,000) number of incoming one-way trips from the originating community (purple triangles in Figure 3a). As a result, we specifically requested all the available samples from the eight communities with the highest incoming mobility. Still, we were restricted to the submission of third parties over whom we had no influence. This led to the inclusion of the following eight communities with the most residential movement from the originating community: four in central and three in NW of Thuringia, one in NW-neighboring state Saxony-Anhalt. The samples requested from central Thuringia were also due to their geographic arrangement as a “belt” in central Thuringia, linking three major cities (see Supplementary Figure S1). Subsequently, we collected 19 additional samples (isolated between the 17th and 25th of October 2022; see “Guided Sampling” for October 2022, Figure 3a) besides the randomized sampling strategy. Thus, the sampling depth was increased in communities with high incoming mobility from the first origin.

As part of the general Thuringian surveillance, we collected 132 samples for October (covering dates between the 5th and 31st) and 69 samples in November (covering dates between the 1st and 25th; see Figure 3b and c). Randomized sampling was not influenced or adjusted based on the mobility-guided sample collection. Thus, it also contains samples from communities with a mobility link towards the first occurrence of BQ.1.1, as they were part of the regular random collection (see gray triangles in Figure 3b). A complete overview of all samples is provided in Supplementary Table S5. The mobility datasets from October 2020 and June 2021 for all sampled communities are provided in Supplementary Table S6.

Line 305 to 313:

Among the 19 samples specifically collected based on mobile service data, we identified one additional sample of the specific Omicron sublineage BQ.1.1 in a community with high incoming mobility ($n = 14$, number of trips = 37,499) with a distance of approximately 16 km between both towns. Our randomly sampled routine surveillance strategy did not detect another sample during the same period. This was despite a seven times higher overall sample rate, which included 31 samples from communities with an identified incoming mobility from the community of the first occurrence (October 2022, Figure 3b). Only in the one-month follow-up were four other samples identified across Thuringia through routine surveillance (November 2022, Figure 3c).

Line 325 to 333:

In summary, increasing the sampling depth in the suspected regions successfully identified the specified lineage using only a fraction of the samples from the randomized sampling. Conversely, randomized surveillance, the “gold standard” acting as our negative control, did not identify additional samples with similar sampling depths in regions with no or low incoming mobility or even in high mobility regions with less sampling depth. Implementing such an approach effectively under pandemic conditions poses difficult challenges due to the fluctuating sampling sizes. Although the finding of the sample may have been coincidental, our proof of concept demonstrated how we can leverage the potential of mobile service data for targeted surveillance sampling.

(2) Line 313: While this work has reliably shown that the spread of Alpha was slower in Thuringia, I don't think there have been sufficient analyses to conclude that this is due to

the lack of transportation hubs. My understanding is that only mobility within Thuringia has been evaluated here and not between Thuringia and other parts of Germany.

Thank you for pointing this out. We noticed that the original sentence lacked the necessary clarity. The statement in line 313 was based on the observation that Alpha first occurred in federal states with major transport hubs, such as international airports and ports, which Thuringia lacks, as demonstrated in the Microreact dataset. For clarification, we adjusted the sentence as follows:

Line 340 and following:

A plausible explanation for the delayed spread of the Alpha lineage in Thuringia is the lack of major transport hubs, as Alpha first occurred in federal states with such hubs. Previous studies have already highlighted the impact of major transportation hubs in the spread of Sars-CoV-2.

(3) Line 333 (and elsewhere): I'm not convinced, based on the results presented in Figure 2, that the authors have reliably identified a sampling bias here. This is only true if you assume (as in line 235) that the variant was in these districts, but that hasn't actually been demonstrated here. While I recognize that for high-prevalence variants, there is a strong correlation between inflow and variant prevalence, low-prevalence variants by definition spread less and may genuinely be missing from some districts. To support this conclusion that they identified a bias, I'd like to see some type of statistical model that is based e.g. on the number of sequences, prevalence of a given variant in other districts, etc. Alternatively, the language can be softened ("putative sampling bias").

Thank you for addressing this legitimate point of criticism in our interpretation. Due to the retrospective nature of the analysis and the fact that we found no additional samples of the clusters after the specified timeframes, we were limited to the samples in our dataset. Therefore, it is impossible to demonstrate if a variant was present in the relevant districts afterward. We agree that the variant's low prevalence means they may genuinely not have spread to some districts. For clarification, we added the following statements and changed the wording accordingly:

Additional statement in line 248:

However, due to their low prevalence, it is also possible that these subclusters have not spread to the indicated districts.

Adjusted wording in line 361:

We exemplified this approach with the Alpha lineage, where mobile service data indicated a putative sampling bias and partially predicted the spread of our Thuringian subclusters.

Recommendations:

(1) I applaud the use of the microreact page to make the data public, however, I don't see any reference to a GitHub or Zenodo repository with the analysis code. The NextStrain code is certainly appreciated but there is presumably additional code used to identify the clusters, generate figures, etc. I generally prefer this code be made public and it is recommended by eLife.

Thank you for your appreciation. We have now included the R-scripts in the manuscript's OSF repository. These were used to create the figures in the manuscript and supplement utilizing the supplementary tables 1-6, which are also stored in the repository. To clearly communicate

which data is provided, we changed lines 513 and 514 of the “Data sharing statement” as follows:

Line 513 and following:

Supplementary tables and the R-scripts used to generate all figures are also provided in the repository under <https://osf.io/n5qj6/>. These include the mobile service data used in this study, which is available in processed and anonymized form.

The subcluster identification was performed manually. By adding each sample’s mutation profile to the Microreact metadata file, we visually screened the phylogenetic time tree for all non-Alpha specific mutations present in at least 20 Thuringian genomes. We then applied the criteria described in the Methods section to identify the nine Alpha subclusters. For clarification, we changed line 436:

Line 436:

We then manually screened for mutations present in at least 20 genomes with a small phylogenetic distance and a time occurrence of at least two months.

Reviewer 2 (Public Review):

In the manuscript, the authors combine SARS-CoV-2 sequence data from a state in Germany and mobility data to help in understanding the movement of the virus and the potential to help decide where to focus sequencing. The global expansion in sequencing capability is a key outcome of the public health response. However, there remains uncertainty about how to maximise the insights the sequence data can give. Improved ability to predict the movement of emergent variants would be a useful public health outcome. Also knowing where to focus sequencing to maximising insights is also key. The presented case study from one State in Germany is therefore a useful addition to the literature. Nevertheless, I have a few comments.

Thank you for taking the time to review our work.

(1) One of the key goals of the paper is to explore whether mobile phone data can help predict the spread of lineages. However, it appears unclear whether this was actually addressed in the analyses. To do this, the authors could hold out data from a period of time, and see whether they can predict where the variants end up being found.

Based on your feedback, we noticed that the results of the other seven clusters presented in the supplement were not appropriately highlighted, causing them to be overlooked. We indeed demonstrated that predicting viral spread based on mobility data is possible, as shown for the high-prevalence subcluster 7 (Cluster “ORF1b:A520V”, 811 samples). This was briefly mentioned in lines 240-242, but the cluster was only shown in Supplementary Figures S4 and S5. Instead, we focused more on the putative sampling bias that the mobility for low-prevalence subclusters could indicate as an interesting use case of mobility data. This addresses a concrete problem of every surveillance: successfully identifying low-prevalence targets. However, based on your feedback, we revisited Figure 2, adding the plots of the high-prevalence subcluster: “ORF1b:A520V” from Supplementary Figures S4 and S5 while moving the low-prevalence subcluster “S:N185D” from Figure 2 into the Supplementary Figures S4 and S5. Additionally, we changed line 229 to highlight this result properly.

line 229 and following:

The mobile service data-based prediction of a subcluster’s spread aligned well with the subsequent regional coverage of fast-spreading, highly prevalent subclusters, such as

subcluster 7, which covered 811 samples (see Figure 2). In contrast, the predicted spread for the low-prevalence subclusters did not correspond well with the actual occurrence.

(2) The abstract presents the mobility-guided sampling as a success, however, the results provide a much more mixed result. Ultimately, it's unclear what having this strategy really achieved. In a quickly moving pandemic, it is unclear what hunting for extra sequences of a specific, already identified, variant really does. I'm not sure what public health action would result, especially given the variant has already been identified.

Thank you for your critical assessment of the presented results and their interpretation.

Here, we aimed to provide an alternative to the standard randomized surveillance strategy. Through mobility-guided sampling, we sought to increase identification chances while necessitating fewer samples and decreasing costs, ultimately enhancing surveillance efficiency. The Omicron-lineage BQ.1.1 was the perfect example to prove this concept under actual pandemic conditions. Yet, the strategy is not limited to low-prevalence sublineages but can be applied to virtually any surveillance case. However, from your question, we recognize that this conclusion was unclear from the text. Therefore, we adapted the conclusion to better communicate the real implications of our proof of concept. Additionally, we altered line 42 in the abstract for clarification.

However, we did not assess the benefits of surveillance itself, as the German Robert Koch Institute (RKI) already had outlined its importance for tracking different viral variants. This tracking served several reasons, like monitoring vaccine escapism, mutational progress, and assessing available antibodies for treatment.

Line 42:

The latter concept was successfully implemented as a proof-of-concept for a mobility-guided sampling strategy in response to the surveillance of Omicron sublineage BQ.1.1.

Line 364 to 374:

Another approach is actively guiding the sampling process through mobile service data, which we demonstrated with our proof of principle focusing on the Omicron-lineage BQ.1.1 as a real-life example. This approach could allow for a flexible allocation of surveillance resources, enabling adaptation to specific circumstances and increasing sampling depth in regions where a variant is anticipated. By incorporating guided sampling, much fewer resources may be needed for unguided or random sampling, thereby reducing overall surveillance costs.

Additionally, while this approach is particularly useful for identifying low-prevalence variants, it is not limited to such variants. Still, it can provide a guided, more cost-efficient, low-sampling alternative to general randomized surveillance that can also be applied to other viruses or lineages.

(3) Relatedly, it is unclear to me whether simply relying on spatial distance would not be an alternative simpler approach than mobile phone data. From Figure 2, it seems clear that a simple proximity matrix would work well at reconstructing viral flow. The authors could compare the correlation of spatial, spatial proximity, and CDR data.

Thank you for pointing this out. While proximity data might appear to be an obvious choice, it has significant limitations compared to mobility data, especially in the context of our study. Proximity data assumes that spatial distance alone can accurately represent movement patterns, which would only be true in a normally distributed traffic network. Geographic features such as mountains, cities, and highways affect traffic flows, leading to variability

over distance and time, which are beyond the scope of spatial proximity but efficiently captured by mobility data. In Figure 2, we presented a simplified view of the mobility data. Hence, proximity and mobility data appear to provide the same insights. However, as shown in the updated Figure 3, a detailed overview of the available mobility data reveals obvious and non-obvious spatial connections that proximity data can not capture. Incorporating such a level of detail in Figure 2 would have cluttered the figure and reduced its clarity (e.g., adding triangles for each Thuringian community).

While a comparison between proximity data and mobility data would indeed be informative, it is beyond the scope of our current study, as our primary focus was to examine the useability of mobility data in explaining our subcluster's spread in the first place. However, we agree it would be a valuable direction for future research. We summarized our thoughts from above in the following additional sentence:

Line 374:

Pre-generated mobility networks automatically tailored to each state's unique infrastructure and population dynamics could provide better-targeted sampling guidance rather than simple geographical proximity.

Recommendations:

(1) Line 128: What do these percentages mean - the proportion of States with at least one Alpha variant? Please clarify.

We clarified the values at their first appearance in the text:

Line 127:

By March, Alpha had spread to nearly all states and districts (districts are similar to counties or provinces) in Germany (Median: 76.47 % Alpha samples among a federal states total sequenced samples compared to 36.03 % in February, excluding Thuringia) and Thuringia (Median: 85.29 %, up from 50.00 % in February).

(2) Line 134: It's a little strange to compare the dynamics of a state with that of the whole country. For it lagged as compared to all other States?

Line 134: "In summary, the spread of the Alpha lineage in Thuringia lagged roughly two weeks behind the general spread in the rest of Germany but showed similar proportions."

Thank you for the feedback. The statement refers to the comparison of Alpha-lineage proportions across federal states, excluding Thuringia, in lines 118 to 130. To simplify, we collectively referred to these federal states as "Germany" in the text. However, we recognize that this formulation is misleading, so we adjusted line 135 for clarification:

Line 135:

In summary, the spread of the Alpha lineage in Thuringia lagged roughly two weeks behind the general spread of other German federal states but showed similar proportions.

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